Ying-Chun (Thomas) Lee

Seattle, WA | thomas910829@gmail.com | +1 (206) 579-1447 | thomas0829.com | github.com/thomas0829

EDUCATION

University of Washington, Seattle, WA — ***Sep 2025 – Mar 2027 (Expected)***
Master of Science in Electrical and Computer Engineering
- GPA: 3.98/4.0

Yuan Ze University, Taoyuan, Taiwan — ***Sep 2020 – Jun 2024***
Bachelor of Science in Electrical Engineering
- Overall GPA: 3.73/4.0, Major GPA: 3.94/4.0

SKILLS

- **Relevant Coursework:** ROS, Self-Driving Cars, Software Development, Embedded Systems, Smart Systems, Artificial Intelligence, Deep Learning, Computer Vision, Internet of Things, Cloud Computing
- **Programming**: Python, C, C++, Java, TypeScript, React, HTML, CSS, SQL
- **Tools & Frameworks:** ROS 2, NVIDIA Isaac Sim, Omniverse USD, PyTorch, OpenCV, NumPy, PIL, Docker, AWS, GCP

TECHNICAL EXPERIENCE

Software Development Engineer Intern, *Amazon, Seattle, WA* — ***Jun 2026 – Sep 2026***
- Designed and implemented a production risk labeling system spanning Aurora PostgreSQL, Java services on AWS Lambda, REST APIs, and a React and TypeScript UI, replacing schema migrations and code deployments for each new risk category with a self-service workflow that reduced recurring report preparation time by about 50%
- Identified and fixed a cross-scope confidentiality leak and a concurrent write race in the Java backend, deriving authorization from persisted records and making label writes idempotent under UNIQUE and primary key contention through explicit transaction handling, constraint translation, and post-transaction reads
- Built keyset-batched bulk labeling for up to 500 risks per request with partial result handling, and developed the AWS Cloudscape governance UI with filtering, grouping, usage drill downs, change history, and CSV and Excel export hardened against spreadsheet formula injection

Research Collaborator, *PRIOR Team at Allen Institute for AI (Ai2), Seattle, WA* — ***Oct 2025 – May 2026***
- Collaborated with Ai2 PRIOR researchers to develop real-robot policy infrastructure for deployment, teleoperation, evaluation, and demos across MolmoSpaces, MolmoAct2, MolmoB0T, and TOPReward
- Fine-tuned and deployed pi0, pi0.5, pi0-FAST, ACT, SmolVLA, and Diffusion Policy across Franka FR3, LeRobot SO-100 and SO-101, and YAM robots, with joint position and velocity control and real-time FR3 inference
- Built a multithreaded stack for teleoperation, policy inference, synchronized logging, replay, calibration, validation, and data conversion, reducing 100-episode rollout time from about 4 hours to 100 minutes while supporting GELLO and Meta Quest 2 control

Intern, *CHANG CHUN GROUP – Information Center, Taipei, Taiwan* — ***Jul 2025 – Aug 2025***
Project: ALOHA-ViperX 300S Digital Twin and Imitation Learning Platform
- Created an end-to-end imitation-learning training platform for the ALOHA VX300S robotic arm, achieving 100% grasp success in simulation across a 35-65 cm workspace using a ROS 2 and Isaac Sim digital twin with a Transformer-based ACT++ control pipeline
- Developed a three-stage trajectory planner with real-time IK constraint validation, workspace-bound checking, and dual-camera perception for robust simulated grasping
- Automated ACT++ HDF5 dataset generation from simulation rollouts and documented a reproducible Isaac Sim workflow covering robot control, perception, trajectory generation, and dataset export

PUBLICATIONS

- Yejin Kim, Wilbert Pumacay, Omar Rayyan, et al., (**Ying-Chun Lee**), “MolmoSpaces: Large-Scale Open Ecosystem for Robot Manipulation and Navigation,” Robotics: Science and Systems (RSS), 2026. arXiv:2602.11337
- **Ying-Chun Lee**, Chih-Yang Lin, Chia-Chun Hsiao, et al., “Using Deep Learning-Based Methods for Automated Segmentation of Soft Tissues from Shoulder Ultrasound Images,” IEEE Access, Vol. 12, pp. 111,481-111,492, July 2024. DOI: 10.1109/ACCESS.2024.3432691
- Abhay Deshpande, Maya Guru, Rose Hendrix, et al., (**Ying-Chun Lee**), “MolmoB0T: Large-Scale Simulation Enables Zero-Shot Manipulation,” under review at the Conference on Robot Learning (CoRL), 2026. Best Poster Award, IEEE ICRA 2026 Workshop on Synthetic Data for Robot Learning. arXiv:2603.16861
- Haoquan Fang, Jiafei Duan, Donovan Clay, et al., (**Ying-Chun Lee**), “MolmoAct2: Action Reasoning Models for Real-world Deployment,” under review at the Conference on Robot Learning (CoRL), 2026. arXiv:2605.02881
- Shirui Chen, Cole Harrison, **Ying-Chun Lee**, et al., “TOPReward: Token Probabilities as Hidden Zero-Shot Rewards for Robotics,” under review at the Conference on Robot Learning (CoRL), 2026. arXiv:2602.19313